

PRACTICAL WEEK-2

TASK1

Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that 1. Accept a Linux command as input. 2. Create a child process using `fork()`. 3. Execute the command in the child process using an appropriate `exec()` system call. 4. Allow the parent process to wait for the child using `wait ()`. 5. Display the Process ID (PID) of both parent and child processes.

- a) Write a C code to create a process using `fork()` system call.**
- b) Read the user input as shell command.**
- c) Use a system call `exec()` to execute the shell command.**
- d) Use a `wait()` system call for waiting a parent process.**
- e) Display the PIDs of parent and child processes.**

```

#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <string.h>

int main()
{
    char command[100];

    printf("Enter a Linux command: ");
    fgets(command, sizeof(command), stdin);

    command[strcspn(command, "\n")] = '\0';

    pid_t pid = fork();

    if (pid < 0)
    {
        perror("fork failed");
        return 1;
    }

    if (pid == 0)
    {
        printf("\nChild Process\n");
        printf("Parent PID : %d\n", getppid());
        printf("Child PID  : %d\n", getpid());

        printf("\nExecuting command: %s\n", command);

        execlp("sh", "sh", "-c", command, (char *)NULL);

        perror("exec failed");
        exit(1);
    }
}

```

[Read 50 lines]

^G Help	^O Write Out	^W Where Is	^K Cut	^T Execute
^X Exit	^R Read File	^\ Replace	^U Paste	^J Justify

```
laasya@laasya:~$ mkdir practical_week2
laasya@laasya:~$ cd practical_week2
laasya@laasya:~/practical_week2$ pwd
/home/laasya/practical_week2
laasya@laasya:~/practical_week2$ nano command_executor.c
laasya@laasya:~/practical_week2$ ls
command_executor.c
laasya@laasya:~/practical_week2$ gcc command_executor.c -o command_executor
laasya@laasya:~/practical_week2$ ls
command_executor  command_executor.c
laasya@laasya:~/practical_week2$ ./command_executor
Enter a Linux command: ls

Parent Process
Parent PID : 563
Child PID  : 566

Child Process
Parent PID : 563
Child PID  : 566

Executing command: ls
command_executor  command_executor.c

Child process completed.
laasya@laasya:~/practical_week2$
```

TASK2:

Using Linux terminal commands (uname, lscpu, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices

```

laasya@laasya:~/practical_week2$ uname -a
Linux laasya 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 x86_64 x86_64 GNU/Linux
laasya@laasya:~/practical_week2$ lscpu
Architecture:                x86_64
  CPU op-mode(s):            32-bit, 64-bit
  Address sizes:              39 bits physical, 48 bits virtual
  Byte Order:                 Little Endian
CPU(s):                      12
  On-line CPU(s) list:       0-11
Vendor ID:                   GenuineIntel
  Model name:                 13th Gen Intel(R) Core(TM) i5-1334U
    CPU family:               6
    Model:                    186
    Thread(s) per core:       2
    Core(s) per socket:       6
    Socket(s):                1
    Stepping:                  3
    BogoMIPS:                  4991.99
  Flags:                      fpu vme de pse tsc msr pae mce cx8 apic sep mtr
                             r pge mca cmov pat pse36 clflush mmx fxsr sse s
                             se2 ss ht syscall nx pdpe1gb rdtscp lm constant
                             _tsc rep_good nopl xtopology tsc_reliable nonst
                             op_tsc cpuid tsc_known_freq pni pclmulqdq vmx s
                             sse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe p
                             opcnt tsc_deadline_timer aes xsave avx f16c rdr
                             and hypervisor lahf_lm abm 3dnowprefetch ssbd i
                             brs ibpb stibp ibrs_enhanced tpr_shadow ept vpi
                             d ept_ad fsgsbase tsc_adjust bmi1 avx2 smep bmi
                             2 erms invpcid rdseed adx smap clflushopt clwb
                             sha_ni xsaveopt xsavec xgetbv1 xsaves avx_vnni
                             vnmi umip waitpkg gfni vaes vpclmulqdq rdpid mo
                             vdiri movdir64b fsrm md_clear serialize ibt flu
                             sh_lld arch_capabilities

Virtualization features:
  Virtualization:            VT-x

```

```
Hypervisor vendor:      Microsoft
Virtualization type:    full
Caches (sum of all):
  L1d:                   288 KiB (6 instances)
  L1i:                   192 KiB (6 instances)
  L2:                    7.5 MiB (6 instances)
  L3:                   12 MiB (1 instance)
NUMA:
  NUMA node(s):          1
  NUMA node0 CPU(s):     0-11
Vulnerabilities:
  Gather data sampling:   Not affected
  Ghostwrite:             Not affected
  Indirect target selection: Not affected
  Itlb multihit:          Not affected
  L1tf:                   Not affected
  Mds:                    Not affected
  Meltdown:               Not affected
  Mmio stale data:        Not affected
  Old microcode:          Not affected
  Reg file data sampling: Mitigation; Clear Register File
  Retbleed:               Mitigation; Enhanced IBRS
  Spec rstack overflow:   Not affected
  Spec store bypass:      Mitigation; Speculative Store Bypass disabled via prctl
  Spectre v1:             Mitigation; usercopy/swapgs barriers and __user pointer sanitization
  Spectre v2:             Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBRSE-eIBRS SW sequence; BHI BHI_DISABLE
  Srbds:                  Not affected
  Tsa:                    Not affected
  Tsx async abort:        Not affected
  Vmscape:                Not affected
laasya@laasya:~/practical_week2$ ps
  PID TTY          TIME CMD
  307 pts/0    00:00:00 bash
  588 pts/0    00:00:00 ps
laasya@laasya:~/practical_week2$ px aux
Command 'px' not found, but can be installed with:
```

```

laasya@laasya:~/practical_week2$ top
top - 05:00:21 up 12 min,  1 user,  load average: 0.00, 0.00, 0.00
Tasks:  22 total,   1 running,  21 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni, 99.9 id,  0.1 wa,  0.0 hi,  0.0 si,  0.
MiB Mem :  7762.6 total,  7217.2 free,   515.5 used,   180.0 buff/cache
MiB Swap:  2048.0 total,  2048.0 free,    0.0 used.  7247.1 avail Mem

```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+
1	root	20	0	21652	13124	9796	S	0.0	0.2	0:00.57
2	root	20	0	3180	2208	2072	S	0.0	0.0	0:00.01
7	root	20	0	3196	2128	2008	S	0.0	0.0	0:00.00
53	root	19	-1	50428	16232	15136	S	0.0	0.2	0:00.16
103	root	20	0	25136	6452	5132	S	0.0	0.1	0:00.31
112	systemd+	20	0	21460	13192	10936	S	0.0	0.2	0:00.09
113	systemd+	20	0	91028	7988	7024	S	0.0	0.1	0:00.06
177	root	20	0	4244	2680	2432	S	0.0	0.0	0:00.01
178	message+	20	0	9596	5344	4660	S	0.0	0.1	0:00.04
191	root	20	0	17968	8840	7796	S	0.0	0.1	0:00.08
194	root	20	0	1755848	13124	10888	S	0.0	0.2	0:00.11
214	syslog	20	0	222516	5956	4552	S	0.0	0.1	0:00.07
216	root	20	0	3124	1972	1836	S	0.0	0.0	0:00.00
231	root	20	0	107020	23192	13728	S	0.0	0.3	0:00.09
302	root	20	0	3188	1112	980	S	0.0	0.0	0:00.00
306	root	20	0	3204	1248	1108	S	0.0	0.0	0:00.06
307	laasya	20	0	6080	5396	3676	S	0.0	0.1	0:00.08
309	root	20	0	6700	4640	3860	S	0.0	0.1	0:00.01
353	laasya	20	0	20316	11644	9516	S	0.0	0.1	0:00.09
354	laasya	20	0	21156	3580	1852	S	0.0	0.0	0:00.00
379	laasya	20	0	6080	5336	3676	S	0.0	0.1	0:00.02
601	laasya	20	0	9300	5680	3496	R	0.0	0.1	0:00.01